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Derive exams

Time response of a II order system for a unity step I/p with $\xi < 1$

KKT transfer function of a II order function

$$\frac{C(s)}{R(s)} = \frac{K\omega_n^2}{\cancel{s(s+2}\ s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$C(s) = R(s) \times \frac{1}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Since the I/p to the system is unit step function then $R(s) = 1/s$

$$C(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

By partial expansion

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\xi\omega_n s + \omega_n^2)}$$

$$\frac{\omega_n^2}{s(s^2 + 2\xi\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\xi\omega_n s + \omega_n^2} \rightarrow (1).$$

$$\frac{\omega_n^2}{s(s^2+2\xi\omega_n s+\omega_n^2)} = \frac{A(s^2+2\xi\omega_n s+\omega_n^2)+B(s)+C}{s(s^2+2\xi\omega_n s+\omega_n^2)}$$

$$\omega_n^2 = A(s^2+2\xi\omega_n s+\omega_n^2 + \cancel{B(s)}\ (Bs+C)s \rightarrow (2)$$

To find find value of A, B, C

Compare the eqn (2) both sides with constant terms

$$\omega_n^2 = A\omega_n^2$$

$$A = \frac{\omega_n^2}{\omega_n^2}$$

$$\boxed{A=1}$$

Equating s square terms on both side of equation (2), we get

$$0 = A+B$$

$$0 = 1+B$$

$$\boxed{B=-1}$$

Equating s terms on both side of eqn (2)

$$0 = A2\xi\omega_n + C$$

$$0 \quad 2\xi\omega_n + C$$

$$\boxed{C = -2\xi\omega_n}$$

Substituting values of A, B & C in eqn (1), we get

$$C(s) = \frac{1}{s} - \frac{s+2\xi\omega_n}{s^2+2\xi\omega_n s+\omega_n^2} \rightarrow (3)$$

Add and subtract $(\xi\omega_n)^2$ in the denominator of ③.

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{\underbrace{s^2 + 2\xi\omega_n s + (\xi\omega_n)^2}_{(a+b)^2} + \omega_n^2 - (\xi\omega_n)^2}$$

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s+\xi\omega_n)^2 + \omega_n^2 - (\xi\omega_n)^2}$$

by taking ω_n^2 common.

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s+\xi\omega_n)^2 + \omega_n^2(1-\xi^2)}$$

Put putting $\omega_n^2(1-\xi^2) = \omega_d^2$

$$C(s) = \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s+\xi\omega_n)^2 + \omega_d^2}$$

$$C(s) = \frac{1}{s} - \frac{s + \xi\omega_n}{(s+\xi\omega_n)^2 + \omega_d^2} + \underbrace{\frac{\xi\omega_n}{(s+\xi\omega_n)^2 + \omega_d^2}}_{*}$$

Divide and multiply * term by ω_d^2.

$$C(s) = \frac{1}{s} - \frac{s + \xi\omega_n}{(s+\xi\omega_n)^2 + \omega_d^2} + \frac{\xi\omega_n}{\omega_d^2} \times \frac{\omega_d^2}{(s+\xi\omega_n)^2 + \omega_d^2}$$

Take ILT

$$C(t) = 1 - e^{-\xi\omega_n t}\left(\cos\omega_d t + \frac{\xi\omega_n}{\omega_d^2} e^{-\xi\omega_n t} \sin\omega_d t\right)$$

$$\omega_d^2 = \omega_n^2(1-\xi^2)$$

$$\omega_d = \sqrt{\omega_n^2} \times \sqrt{(1-\xi^2)}$$

$$\omega_d = \omega_n\left(\sqrt{1-\xi^2}\right)$$

$$C(s) = \frac{1}{s} - \left[\frac{s+\xi\omega_n}{(s+\xi\omega_n)^2} + \frac{\xi\omega_n}{\omega_n\sqrt{1-\xi^2}} \left[\frac{\omega_d^2}{(s+\xi\omega_n)^2+\omega_d^2} \right] \right]$$

$$C(s) = \frac{1}{s} - \left[\frac{s+\xi\omega_n}{(s+\xi\omega_n)^2+\omega_d^2} + \frac{\xi}{\sqrt{1-\xi^2}} \times \frac{\omega_d^2}{(s+\xi\omega_n)^2+\omega_d^2} \right]$$

Take inverse laplace transform.

$$c(t) = 1 - \left[e^{-\xi\omega_n t}\cos\omega_n t + \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin\omega_n t \right]$$

By taking $e^{-\xi\omega_n}$ common from the above eqn.

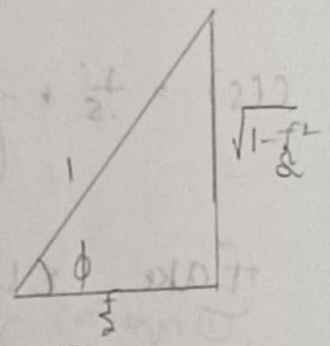
$$c(t) = 1 - e^{-\xi\omega_n t}\left(\cos\omega_n t + \frac{\xi}{\sqrt{1-\xi^2}}\sin\omega_n t\right)$$

By taking $\frac{1}{\sqrt{1-\xi^2}}$ as common.

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sqrt{1-\xi}\cos\omega_n t + \xi\sin\omega_n t\right]$$

$$\sin\phi = \frac{\text{opp}}{\text{HYP}} = \frac{\sqrt{1-\xi^2}}{1} = \sqrt{1-\xi^2}$$

$$\cos\phi = \frac{\text{Adj}}{\text{HYP}} = \frac{\xi}{1} = \xi$$



$c(t) = [\sin(a+b) = \sin a\cos b + \cos a\sin b]$

$\sin(\omega_d t + \phi) = \sin\omega_d t\cos\phi + \cos\omega_d t\sin\phi$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sqrt{1-\xi}\cos\omega_n t + \xi\sin\omega_n t\right]$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sin\omega_d t + \phi\right]$$

Time response of a II order system with unit step input and damping value i.e. ξ value = 1.

K/KT,

$$C(s) = R(s) \times \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \rightarrow (1)$$

$$C(s) = \frac{1}{s} \times \frac{\omega_n^2}{\underbrace{s^2 + 2\omega_n s + \omega_n^2}_{(a+b)^2}}$$

$R(s) = 1/s$
$\xi = 1$

$$= \frac{1}{s} \times \frac{\omega_n^2}{(s+\omega_n)^2}$$

$$= \frac{1}{s} \times \omega_n^2 \times \frac{1}{(s+\omega_n)^2}$$

$$C(s) = \frac{1}{s} \times \frac{\omega_n^2}{(s+\omega_n)^2}$$

$$C(s) = \frac{\omega_n^2}{s(s+\omega_n)^2}$$

(By partial fraction method)

$$C(s) = \frac{A}{s} + \frac{Bs+C}{(s+\omega_n)^2}$$

$$\frac{\omega_n^2}{s(s+\omega_n)^2} = \frac{A}{s} + \frac{Bs+C}{(s+\omega_n)^2}$$

$$\frac{\omega_n^2}{\cancel{s(s+\omega_n)^2}} = \frac{A(s+\omega_n)^2 + Bs + C(s)}{\cancel{s(s+\omega_n)^2}}$$

Equating ω_n^2 on either side of equation

$$C(s) = \frac{1}{s} \frac{\omega_n}{(s+\omega_n)} \times \frac{\omega_n}{(s+\omega_n)}$$

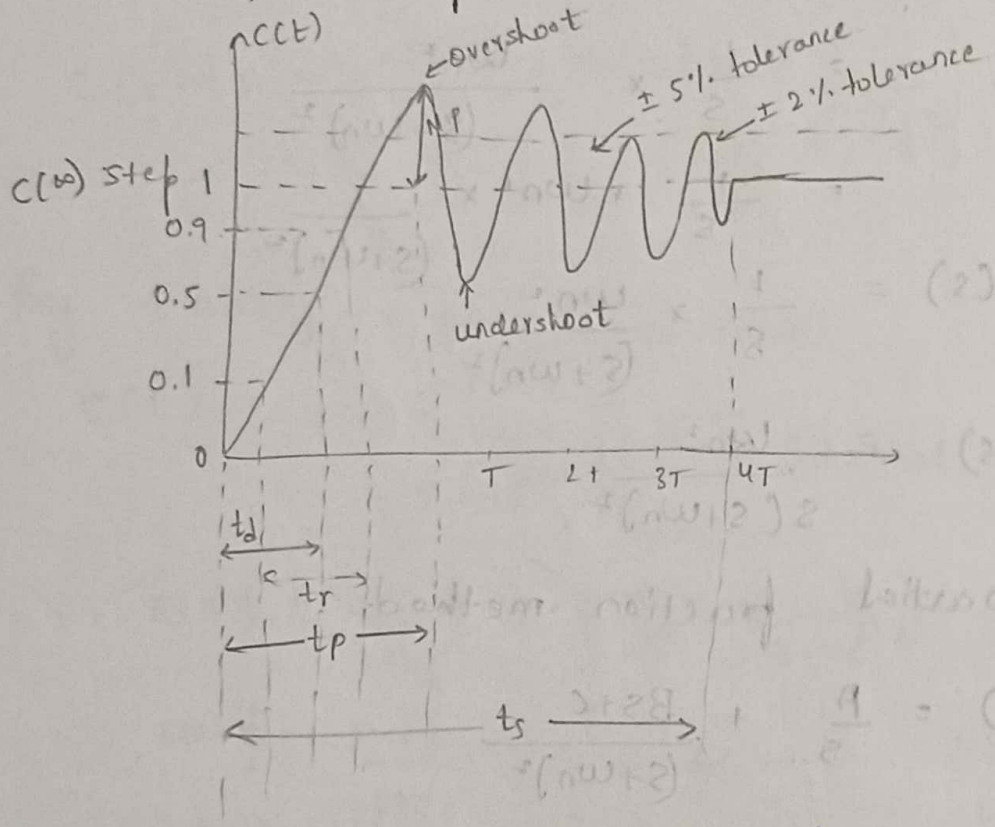
$$C(s) = \omega_n \left[\omega_n \frac{1}{s} \times \frac{1}{s+\omega_n} \times \frac{1}{s+\omega_n} \right]$$

Take ILT:

$$c(t) = \omega_n \left[\omega_n \left(1 - e^{-\omega_n t} \times e^{-\omega_n t}\right) \right]$$

$$\boxed{c(t) = \omega_n^2 \left[1 - e^{(-\omega_n t)^2} \right]}$$ when $\omega_n \geq 1$.

Time response specification for II order system.



t_d = delay time → It is that part of the response by which it takes 50% of it's maximum final value.

time taken by response to reach 90% of it's maximum value

t_p = peak time: time taken by the response to reach above the maximum value.

It is ~~defined~~ as.

Delay time 'td': It is defined as the time taken by response to reach 50% of its final value for the first time.

rise time (tr): It is defined as the time taken by the response to reach 90% of its final value from 10% of its initial value for the first time

Peak time: (tp): It is the time required to the response to reach maximum (or) peak value of first overshoot

Settling time (ts) Time taken by the response to reach and stay within a tolerance band of ±5% (or) ±2% of the steady state value.

Maximum (or) peak overshoot (Mp):

It is the maximum (or) peak value of response curve measured from step unity.

It is mathematically given by.

$$\% M_p = \frac{C(t_p) - C(\infty)}{C(\infty)} \times 100$$

Other than these parameters we may come across with ω_n natural frequency (or) undamped frequency, ω_d: damped frequency.

damping ratio (ξ).

All these above parameters are represented by the mathematical equations as below.

$$t_p = \frac{n\pi}{\omega_d}$$

where n = 0, 1, 2, 3 etc.

If $n=0$: $t_p = 0$ it is called as zero order overshoot.

If $n=1$, $t_p = \frac{\pi}{\omega_d}$ is called as first order overshoot.

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} \longrightarrow (1)$$

$$\omega_d^2 = \omega_n^2(1-\xi^2)$$

$$t_r = \frac{\pi - \phi}{\omega_d} \text{ sec} \longrightarrow (2)$$

$$\omega_d = \sqrt{\omega_n^2}\sqrt{1-\xi^2}$$

$$\omega_d = \omega_n\sqrt{1-\xi^2}$$

$$\% M_p = e^{-\pi\xi/\sqrt{1-\xi}} \times 100 \longrightarrow (3)$$

$$t_s = e^{-\xi\omega_n t_s} \longrightarrow (4)$$

① Evaluate the following quantities for a second order system with open loop transfer function $G(s) = \frac{25}{s(s+7)}$ for a unit step input.

(i) Undamped natural frequency (ω_n) in radian

(ii) time ξ (Xita)

(iii) time constant 'T'

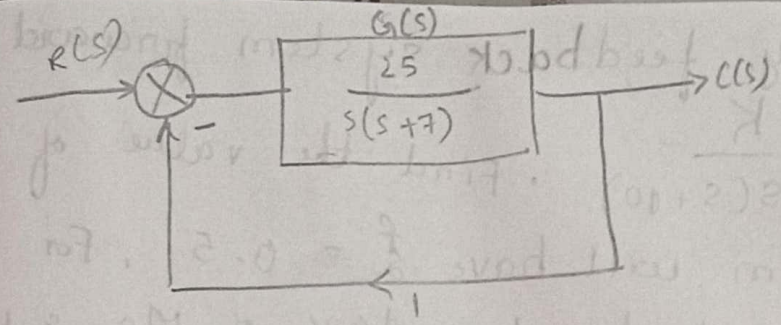
(iv) damped natural frequency ω_d

(v) rise time (t_r)

(vi) peak time (t_p)

(vii) setting time (t_s)

(viii) percentage overshoot (M_p)



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$= \frac{25/s(s+7)}{1+\frac{25}{s(s+7)}(1)}$$

$$= \frac{\frac{25}{s(s+7)}}{\frac{s(s+7)+25}{s(s+7)}}$$

$$= \frac{25}{s^2+7s+25} \longrightarrow (1)$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \longrightarrow (2)$$

$\omega_n^2 = 25 \quad \therefore \omega_n = \sqrt{25} = 5$ radian.

$2\xi\omega_n = 7$

$\xi = 7/2\omega_n = 7/5\times 2 = 7/10 = 0.7$

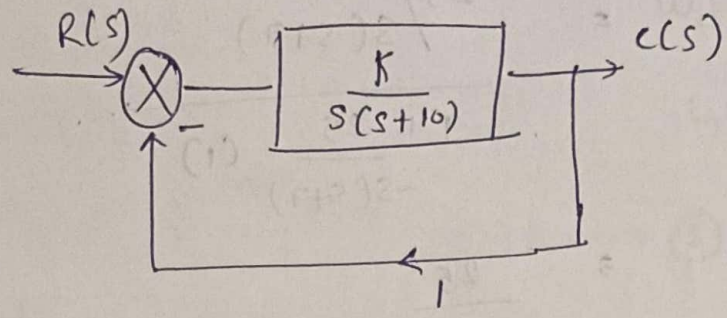
$$T = \frac{1}{\xi\omega_n}$$

$$= \frac{1}{0.7\times 5}$$

$T = 0.28$ sec

① A unity feedback system forward path gain $C(s) = \frac{K}{s(s+10)}$. Find the value of K so that the system will have $\xi = 0.5$. For this value of K find percentage of M_p & t_p.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$



$$= \frac{K/s(s+10)}{1+\frac{K}{s(s+10)} \times 1}$$

$$= \frac{\frac{K}{s(s+10)}}{\frac{s(s+10)+K}{s(s+10)}}$$

$$\frac{C(s)}{R(s)} = \frac{K}{s(s+10)+K} \longrightarrow ①$$

WKT, standard equation for closed loop response of a II order system.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \longrightarrow ②$$

comparing eqn ① & ②

$\omega_n^2 = K$

$\omega_n = \sqrt{K}$

$2\xi\omega_n = 10$

$\omega_n = \frac{10}{2\xi}$

$= \frac{10}{2(0.5)}$

$\boxed{\omega_n = 10 \text{ rad/sec}}$

ω_n = natural undamped frequency.

$\% M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100$

$= e^{-\frac{0.5\times\pi}{\sqrt{1-0.5^2}}} \times 100$

$\% M_p = 16.3\%$

$t_p = \pi/\omega_d = \frac{\pi}{8.66} = 0.36$ sec

WKT,

$\omega_d = \omega_n\sqrt{1-\xi^2}$

$= 10\sqrt{1-(0.5)^2}$

$\boxed{\omega_d = 8.66 \text{ rad/sec}}$

I order $t_p = \frac{\pi}{\omega_d}$

II order $t_p = \frac{n\pi}{\omega_d}$

(03) Time response of a II order system for unit step input is $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$. Determine

(i) Closed loop transfer function.

(ii) value of ω_n and damping ratio.

(iii) justify the system is under damped or over damped

: Since the given eqn for ~~or~~ expression is in the form of time response which is to be converted into laplace domain.

$c(t) = 1 + 0.2e^{-60t} + 1.2e^{-10t}$

Apply L-T

$C(s) = \frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{s+10}$

$$= \frac{1(s+60)(s+10) + 0.2s(s+10) - 1.2s(s+60)}{s(s+60)(s+10)}$$

$$= \frac{s^2 + 10s + 60s + 600 + 0.2s^2 + 2s - 1.2s^2 - 72s}{s(s^2+10s+60s+600)}$$

$$C(s) = \frac{1}{s}\frac{600}{(s^2+70s+600)}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{600}{s^2+70s+600}} \rightarrow (1)$$

WKT, standard equation for II order system, closed loop form is given by

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \longrightarrow (2)$$

Comparing (1) & (2)

$\omega_n^2 = 600$

$\omega_n = \sqrt{600}$

$\boxed{\omega_n = 24.49}$

$2\xi\omega_n = 70 \Rightarrow \xi = \frac{70}{2\times\omega_n}$

$\cancel{\omega_n = \frac{70}{2\xi}}$

$= \frac{70}{2\times 24.49}$

$= \frac{70}{2\times 1.42}$

$\boxed{\xi = 1.42}$

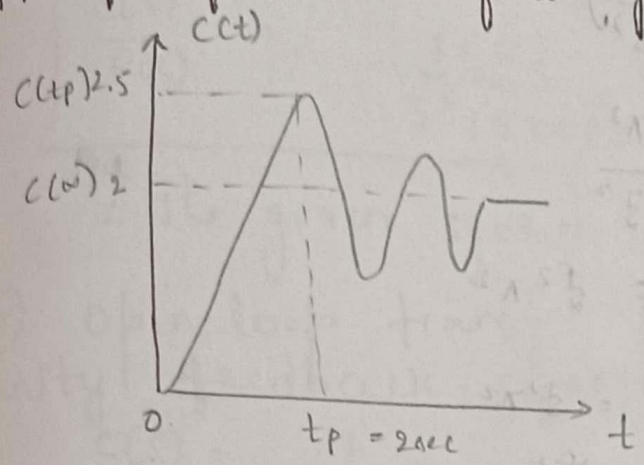
$\% M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100$

$= e^{-1.42\times\pi/\sqrt{1-1.42}} \times 100$

$\% M_p =$

Since $\xi = 1.42$ which is greater than 1 hence it is overdamped system.

(09) Time response of II order under damped unity feedback system applied with u(t) as I/P shown in figure below. Find the closed loop transfer function of a system



: Given,

$$\% M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100$$

$$= \frac{2.5 \times 2}{2} \times 100$$

$$\boxed{\% M_p = 25\%}$$

$$t_p = \frac{n\pi}{\omega_d}$$

$$= \frac{2 \times \pi}{\omega_d}$$

$$= \frac{\cancel{2} \times \pi}{\omega_d \cancel{2}}$$

$$\boxed{t_p = \pi = 3.142}$$

$$M_p = e^{-\xi\pi / \sqrt{1-\xi^2}}$$

$$0.25 = e^{-\xi\pi / \sqrt{1-\xi^2}}$$

$$\ln 0.25 = -\xi\pi / \sqrt{1-\xi^2}$$

$$-1.38 = \frac{\xi\pi}{\sqrt{1-\xi^2}}$$

Squaring on both side

$$(-1.38)^2 = \left(\frac{\xi\pi}{\sqrt{1-\xi^2}}\right)^2$$

$$1.92 = \frac{\xi^2\pi^2}{1-\xi^2}$$

$$1.92(1-\xi^2) = \xi^2\pi^2$$

$$1.92 - 1.92\xi^2 = \xi^2\pi^2$$

$$\xi^2\pi^2 + 1.92\xi^2 = 1.92$$

$$\xi^2(\pi^2 + 1.92) = 1.92$$

$$\xi^2 = \frac{1.92}{\pi^2 + 1.92}$$

$$\xi^2 = 0.162$$

$$\xi = \sqrt{0.162}$$

$$\boxed{\xi = 0.402}$$

$$t_p = \frac{\pi}{\omega_d}$$

$$\omega_d = \frac{2\pi}{t_p} = \frac{2\pi}{t_p} = \pi$$

$$\omega_d = \omega_n\sqrt{1-\xi}$$

$$\omega_n = \frac{\omega_d}{\sqrt{1-\xi^2}}$$

$$= \frac{\pi}{\sqrt{1-0.402}} \Rightarrow \boxed{\omega_n = 3.433}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$= \frac{(3.43)^2}{s^2 + (2 \times 0.402 \times 3.43)s + (3.43)^2}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{11.76}{s^2 + 2.75s + 11.76}}$$ → Closed loop transfer func.

∴ The given system is underdamped.

(05) Open loop transfer function of a unity feedback system is given by

$$C(s) = \frac{1}{s^2T + s + K}$$

(i) By what factor value of K should be multiplied so that the damping ratio is in the 0.2 to 0.8.

(ii) By what factor the value of 'K' should be multiplied so that the system overshoot is ~~increased from 16%~~ reduced from 60% to 20% for a unit step input.

∴ Since the given equation is not in the standard form hence dividing both numerator & denominator by 'T'

$$\frac{C(s)}{R(s)} = \frac{K/T}{s^2 + \frac{s}{T} + \frac{K}{T}}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$\omega_n^2 = K/T$

$\omega_n = \sqrt{K/T}$

$2\xi\omega_n = 1/T$

$2\xi\sqrt{K/T} = 1/T$

$\xi = \dfrac{1}{T\,2\sqrt{K/T}}$

Squaring on B.S

$(\xi)^2 = \left(\dfrac{1}{T\,2\sqrt{K/T}}\right)^2$

$\xi^2 = \dfrac{1}{T^2\,4K/T}$

$\Rightarrow \xi = \sqrt{\dfrac{1}{4TK}}$

$\cancel{\xi = \sqrt{\dfrac{1}{2\sqrt{KT}}}}$

$\boxed{\xi = \dfrac{1}{2}\dfrac{1}{\sqrt{TK}}}$

for $\xi = 0.2$

$2\xi\omega_n = \dfrac{1}{T}$

$2(0.2)\sqrt{\dfrac{K}{T}} = \dfrac{1}{T}$

Squaring on both side

$(0.4)^2\dfrac{K}{\cancel{T}} = \dfrac{1}{T^{\cancel{2}}}$

$K = \dfrac{1}{0.16T}$ or $\boxed{K = \dfrac{6.25}{T}}$

for $\xi = 0.8$

$2\xi\omega_n = \dfrac{1}{T}$

$2(0.8)\sqrt{\dfrac{K}{T}} = \dfrac{1}{T}$

$(1.6)^2\dfrac{K}{\cancel{T}} = \dfrac{1}{T^{\cancel{2}}}$

$K = \dfrac{1}{(1.6)^2 T}$

$K = \dfrac{1}{2.56T}$ $\boxed{K = \dfrac{0.39}{T}}$